

KHAMIS MUSHAIT, Saudi Arabia

WMO: 411140

Lat: 18.297N

Lon: 42.804E

Elev: 6778

StdP: 11.44

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 45.5	(c) 47.9	(d) 17.5	(e) 17.1	(f) 65.0	(g) 21.5	(h) 20.7	(i) 67.6	(j) 23.3	(k) 64.0	(l) 21.6	(m) 64.5	(n) 1.7	(o) 180	(p) 0.391

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 22.1	(c) 90.7	(d) 58.5	(e) 89.4	(f) 58.9	(g) 87.9	(h) 59.1	(i) 67.5	(j) 76.7	(k) 66.4	(l) 75.6	(m) 65.4	(n) 74.8	(o) 12.5	(p) 60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	118.2	72.1	63.8	114.7	71.7	62.5	109.1	71.1	36.5	77.5	35.4	74.8	34.6	74.7	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 21.5	(b) 19.2	(c) 17.1	(e) 40.4	(f) 93.6	(g) 2.7	(h) 2.9	(i) 38.4	(j) 95.7	(k) 36.8	(l) 97.4	(m) 35.2	(n) 99.0	(o) 33.2	(p) 101.1		
			DB													
			WB													
				32.9	70.6	3.2	2.3	30.7	72.3	28.8	73.7	27.0	75.0	24.7	76.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	69.1	59.4	62.5	65.4	69.1	73.9	77.3	77.4	76.7	74.6	68.7	63.7	60.2		
	DBStd	7.02	3.02	3.05	3.16	3.12	3.05	2.02	2.50	2.13	2.21	2.40	2.75	2.59		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	499	174	79	33	6	1	0	0	0	0	2	56	148		
	CDD50	6974	292	351	478	574	741	818	848	826	738	579	412	317		
	CDD65	1998	1	10	46	130	277	368	383	361	288	116	17	1		
	CDH74	16648	50	120	299	841	2231	3440	3224	2947	2482	813	160	41		
	CDH80	5579	1	3	14	122	729	1415	1262	1165	792	73	3	0		
(14) Wind	WSAvg	7.0	7.0	7.6	8.4	7.4	6.4	7.1	7.3	7.1	7.7	6.9	5.4	5.6		
(15) Precipitation	PrecAvg	7.0	0.3	0.2	0.9	1.6	1.2	0.4	0.8	1.0	0.2	0.2	0.2	0.0		
	PrecMax	16.4	1.5	2.2	9.8	7.6	3.1	3.2	2.8	5.5	1.6	1.5	1.2	0.5		
	PrecMin	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0		
	PrecStd	3.7	0.4	0.5	1.9	1.9	0.9	0.7	0.7	1.1	0.3	0.4	0.3	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.4	80.3	82.0	85.8	89.9	93.1	93.4	91.6	90.0	84.0	80.2	77.3		
		MCWB	54.1	56.7	57.7	58.2	59.1	57.1	58.1	59.1	57.6	55.1	55.7	53.5		
	2%	DB	75.2	77.3	79.2	82.8	88.0	90.0	90.4	89.6	87.9	82.2	77.4	75.0		
		MCWB	52.9	56.0	56.9	58.5	58.8	58.2	58.8	59.8	58.2	55.3	55.5	54.2		
	5%	DB	72.0	75.2	77.2	80.9	86.2	88.8	88.2	88.0	86.2	80.5	75.6	73.0		
		MCWB	53.2	55.1	56.9	58.4	58.6	58.4	59.3	59.9	57.8	54.7	55.4	54.4		
	10%	DB	69.7	73.1	75.3	79.0	84.4	87.5	86.4	86.3	84.4	78.7	73.7	70.2		
		MCWB	53.3	55.3	57.0	58.7	58.8	58.4	59.3	59.7	57.4	54.7	55.2	54.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.9	62.8	63.7	65.4	66.9	67.9	68.9	69.0	66.8	63.0	62.6	61.2		
		MCDB	68.6	74.4	73.8	77.1	76.6	77.8	78.6	79.1	75.1	70.4	71.5	69.7		
	2%	WB	58.0	60.3	61.6	63.3	65.3	66.2	67.0	67.5	65.1	61.4	60.9	59.4		
		MCDB	66.7	70.7	72.4	74.7	75.6	76.6	76.4	76.5	73.6	69.3	69.0	67.9		
	5%	WB	56.8	58.9	60.4	62.3	64.0	64.7	65.7	66.4	63.7	60.2	59.9	58.0		
		MCDB	65.3	68.9	70.9	73.7	75.0	75.7	75.3	75.0	73.0	69.0	68.0	66.5		
	10%	WB	55.8	57.6	59.3	60.9	62.7	63.2	64.5	65.3	62.0	58.8	58.7	56.9		
		MCDB	64.5	67.2	69.8	72.6	74.9	75.7	74.7	74.0	73.6	69.2	67.0	65.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	20.9	21.0	19.5	20.3	22.1	22.1	20.9	22.0	22.0	21.8	21.8	22.2		
		MCDBR	26.2	24.8	22.7	22.5	23.4	23.4	22.3	23.1	23.2	22.9	23.7	26.0		
		MCWBR	13.4	12.6	11.3	10.3	10.0	10.4	9.7	9.9	10.5	11.4	11.8	13.0		
	5% WB	MCDBR	20.8	21.5	19.4	20.5	22.5	22.4	21.4	22.1	22.4	22.0	21.3	21.3		
		MCWBR	11.5	12.1	10.5	10.2	10.4	11.3	10.3	10.4	11.8	12.5	11.2	11.7		
(40) Clear-Sky Solar Irradiance	taub		0.327	0.365	0.403	0.450	0.508	0.564	0.624	0.610	0.502	0.390	0.369	0.333		
	taud		2.412	2.265	2.144	2.019	1.844	1.690	1.611	1.635	1.837	2.157	2.248	2.378		
	Ebn at Noon		304	296	287	273	255	239	225	230	255	284	286	298		
	Edh at Noon		35	43	50	57	67	77	84	82	67	47	41	35		
(44) All-Sky Solar Radiation	RadAvg		1625	1861	2093	2184	2281	2300	2090	2017	2089	2025	1716	1588		
	RadStd		82	91	98	122	117	79	88	67	64	69	60	85		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.65	N/A	N/A	N/A	N/A	N/A	N/A	-61	+232	+178
(47) Regional (4 neighbors)		+0.85	N/A	N/A	+0.88	N/A	N/A	N/A	-20	+346	+277

Nomenclature: See separate page